Hongyi Ding

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EDUCATION

Northeastern University, Boston, MA 2023 – Present
Bachelor of Science in Computer Science and Physics, Expected May 2027
Dean's List of Khoury College of Computer Science

INTERNSHIP

ZheJiang Supcon Information Co., Ltd. Jul. 2026 – Nov. 2026
Software Development Intern Hangzhou, China

- Developed and maintained backend modules for a video algorithm platform, contributing to requirement analysis, feature design, and implementation
- Built and deployed core backend functionality using C++ and the Drogon framework, and integrated ZLMediaKit for video proxying and stream distribution
- Deployed and extended the File Browser file-management system on Linux servers using Docker, adding functionality to support subsequent development of the video platform
- Collaborated closely with algorithm and frontend teams, designed API documentation, supported cross-module integration, and optimized application workflows

PROJECTS

Scratch Novel Engine Jun. 2026 – Present
Electron, C++ Personal Project

- Built a low-code visual novel development tool using Electron, React, TypeScript, Blockly, and C++20, supporting form-based, block-based, and DSL-based editing
- Designed an IPC and JSONL-based architecture with a C++ domain model as the single source of truth, ensuring consistent data across multiple editors and safe local file storage
- Incorporated feedback from an ENGW 3302 instructor to improve the tool's usability and narrative authoring workflow through iterative updates

Slay the Spire 2 Character Mod — OTTO Apr. 2026 – Oct. 2026
C#, Godot, BaseLib Personal Project

- Developed a playable character mod using C#, Godot, and BaseLib, implementing a custom starting deck, character-specific cards, relics, and character visuals
- Designed and implemented custom combat mechanics, including character transformation, summonable companions, and card combos, with persistent state across save and load operations
- Integrated custom artwork, animations, voice lines, and English/Chinese localization; published the mod on Steam Workshop with BaseLib dependency support and continue to refine gameplay based on community feedback

F1 Vehicle Dynamics and Aerodynamic Performance Optimization Jan. 2026 – Apr. 2026
Python, Matplotlib Personal Project

- Developed longitudinal and lateral vehicle dynamics models to analyze the effects of aerodynamic drag, downforce, and tire grip on F1 vehicle performance
- Implemented a simplified two-dimensional CFD airflow model based on the Navier–Stokes equations and the Finite Volume Method (FVM)
- Analyzed the C_L–C_D trade-off and optimized aerodynamic configurations for different circuit characteristics using lap-time modeling

Mu's Adventure Oct. 2025 – Present
Unity Team Project

- Developed a 2D side-scrolling platform game in which players solve puzzles using clues distributed throughout the game world
- Developed multiple mini-games and gameplay systems using the Unity engine
- Integrated world-building and gameplay mechanics to create a cohesive narrative and interactive experience

Global Game Jam 2026 Feb. 2026
Unity, Git Team Project

- Developed a playable Unity game within a 48-hour development cycle based on the theme "Mask"
- Designed and integrated mask-related gameplay mechanics, including state transitions and visual feedback systems
- Collaborated in a fast-paced team environment and used Git for version control and iterative development

TECHNICAL SKILLS

Programming Languages: C++, C#, Python
Operating Systems: macOS, Windows, Linux
Development Tools: Visual Studio Code, Git, Codex, Cursor
Frameworks / Technologies: Unity, Godot, Electron, Drogon, Docker, ZLMediaKit
Applications: Microsoft Office, Adobe Photoshop, Adobe Premiere Pro

VOLUNTEER

The Green Lion May 2020 – Jul. 2020
Volunteer Bali, Indonesia

- Delivered foundational courses to local primary school students
- Participated in beach clean-up activities to remove litter and promote environmental conservation
- Assisted in cleaning and maintaining sea turtle habitats to support marine wildlife conservation

Local Community Sep. 2017 – Jun. 2020
Volunteer Suzhou, Jiangsu

- Monitored and managed daily community traffic flow to support operational efficiency
- Tracked the well-being of community households and helped ensure timely support for residents with special needs
- Prepared monthly summaries and recommendations for community improvements

OTHER

- **Portfolio:** darkboyryan.github.io/HongyiDing.github.io
- **GitHub:** github.com/DarkboyRyan
- **Languages:** Mandarin, English